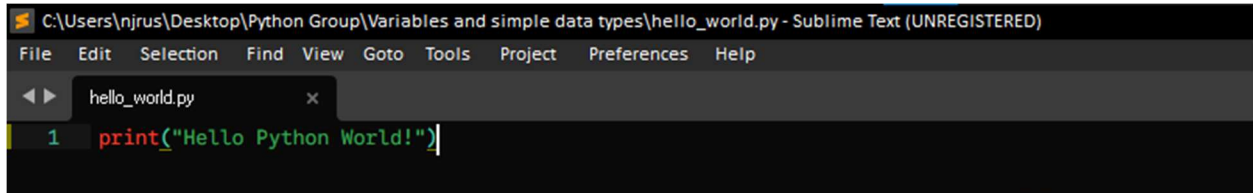


FROM BASICS TO JUNIOR PYTHON DEV: Variables & Data Types (A)

In this lesson, you will learn how to use variables in your programs.

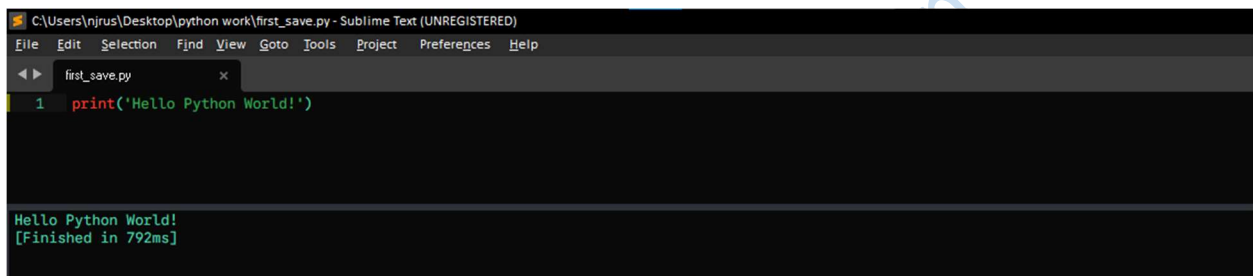
Variables

Create a new file and name it **hello_world.py** inside your project folder. In this new file, write the following line of code.



```
C:\Users\njrus\Desktop\Python Group\Variables and simple data types\hello_world.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
hello_world.py x
1 print("Hello Python World!")
```

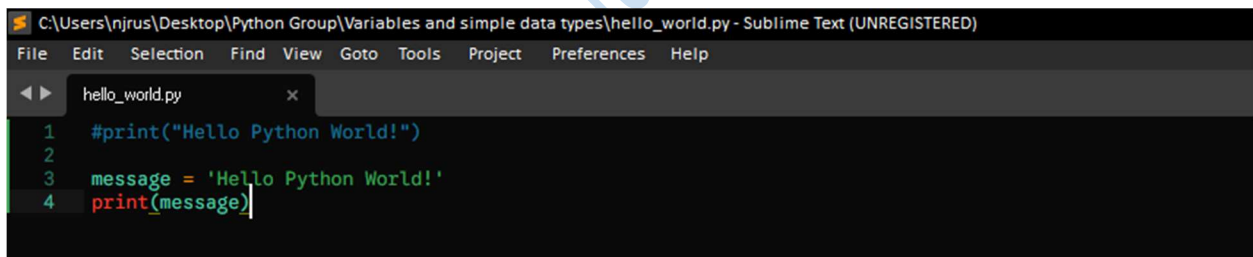
When you run this code, you should see this output:



```
C:\Users\njrus\Desktop\python work\first_save.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
first_save.py x
1 print('Hello Python World!')

Hello Python World!
[Finished in 792ms]
```

Now, comment out that line and add this instead:



```
C:\Users\njrus\Desktop\Python Group\Variables and simple data types\hello_world.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
hello_world.py x
1 #print("Hello Python World!")
2
3 message = 'Hello Python World!'
4 print(message)
```

When you run this program now, you should see the same output as the one you saw earlier. This introduces us to the concept of variables.

Definition:

A **Variable** is a name used to refer to a value stored in a program (a program like `hello_world.py`).

In the above code, the word **message** is a variable and **Hello Python World** is the value associated with the variable `message`.

Creating Variables:

A variable is created when you assign some piece of data to a name using `=`. See below:



FROM BASICS TO JUNIOR PYTHON DEV: Variables & Data Types (A)

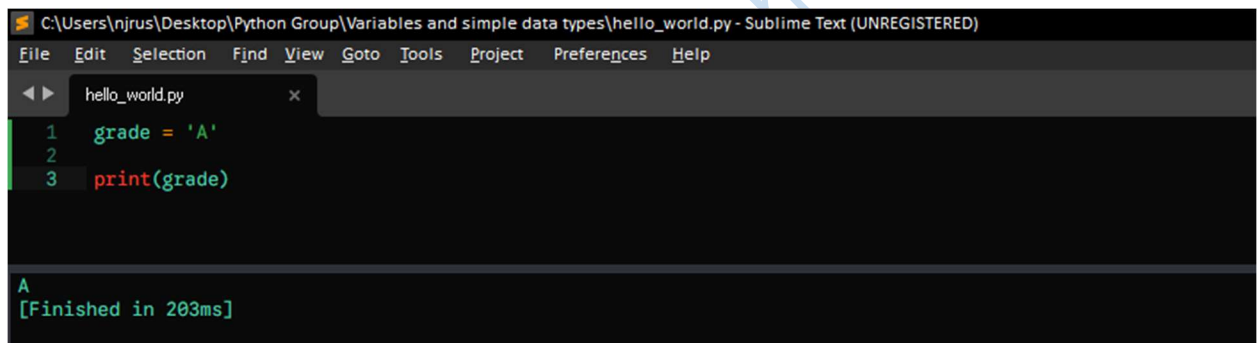
```
3
4 name = 'John'
5 age = 20
6 height = 1.75
7 is_a_student = False
```

Here, '=' is called an **assignment operator** and the process of taking a value and assigning it to a variable is called **assignment**. Through the lifetime of a program, unless the variable's value is changed, whenever you print the value associated with **name**, **age**, **height** and/or **is_a_student**, you will see **John**, **20**, **1.75** and/or **False** respectively as the only outputs.

Some key points to note about python variables:

- **A variable can be reassigned**

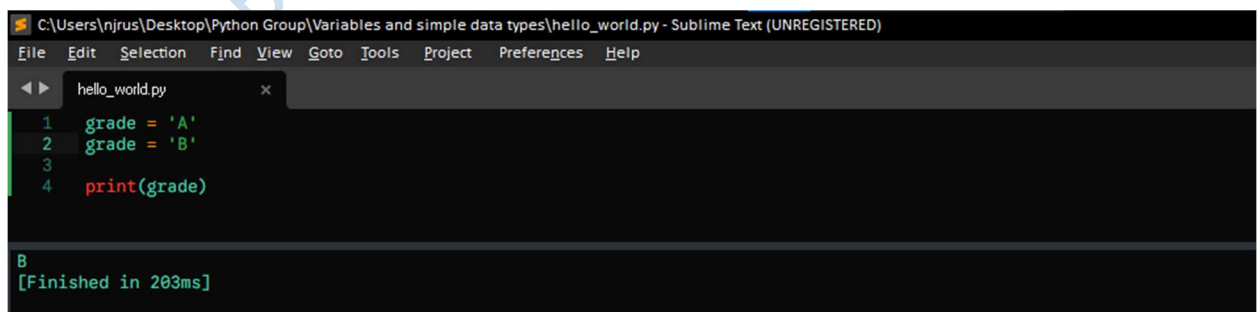
Defining a variable doesn't necessarily mean that its value cannot be changed. Python allows overwriting of a variable's value if needed. See below:



```
C:\Users\njrus\Desktop\Python Group\Variables and simple data types\hello_world.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
hello_world.py x
1 grade = 'A'
2
3 print(grade)

A
[Finished in 203ms]
```

Whenever you reassign a variable to another value, that variable no longer references the original value but rather the value on the latest assignment. See Below:



```
C:\Users\njrus\Desktop\Python Group\Variables and simple data types\hello_world.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
hello_world.py x
1 grade = 'A'
2 grade = 'B'
3
4 print(grade)

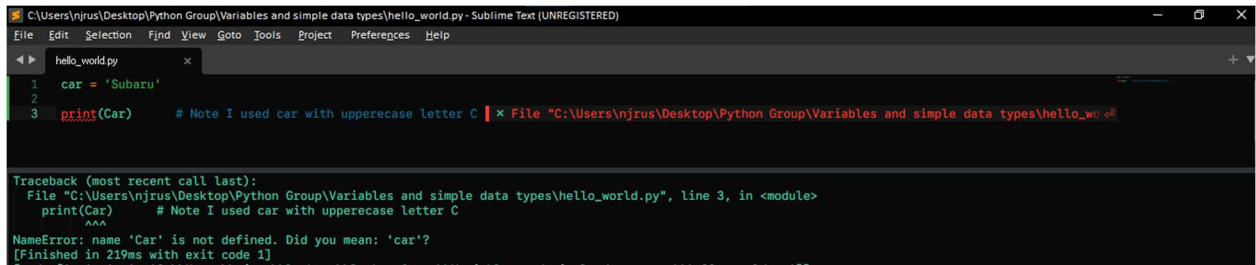
B
[Finished in 203ms]
```

This is called **reassignment**.

- **Variable names are case-sensitive**

Python treats uppercase and lowercase letters different. Therefore, to python, **name** and **Name** are two different variables. Consider the following program:





```
C:\Users\njrus\Desktop\Python Group\Variables and simple data types\hello_world.py - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help
hello_world.py
1 car = 'Subaru'
2
3 print(Car) # Note I used car with upperecase letter C
***
Traceback (most recent call last):
  File "C:\Users\njrus\Desktop\Python Group\Variables and simple data types\hello_world.py", line 3, in <module>
    print(Car)
    ^^^^^
NameError: name 'Car' is not defined. Did you mean: 'car'?
[Finished in 219ms with exit code 1]
```

Python could not print the value associated with Car since the variable Car was not defined. In this case, Python threw a **NameError**, which simply means that you tried to access a variable's value before defining the actual variable.

- **Naming your variables**

Variable names can only contain letters, numbers and underscores. They can start with a letter or an underscore, **BUT NOT with a number**. For instance, you may want to name a variable **message_1** but not **1_message**.

Spaces are not allowed in variable names. Instead, we use underscores to separate words in variable names. For example, **my_name** works in a program but **my name** will cause errors. Therefore, if your variable contain more that one word, ensure you concatenate the variable using underscore(s).

Variable names should be short and descriptive. For example **'name'** is better that **'n'**, **'student_name'** is better that **'s_n'**, **'name_length'** is better than **'length_of_persons_name'**.

It takes practice to learn how to create good variables.

Please find time to go through the assignments below.



Try Yourself

1. Create variables to store and print the following information about yourself:
 - Your name
 - Your age
 - Your country
 - Your height
2. Create two variables that will store the price and quantity bought of a product respectively. Create another variable called total that calculates the total cost. Print the result.
3. Create variables for length and width of a rectangle and use them to calculate the area and the perimeter. Print the results.
Area = length x width Perimeter = 2 x (length + width)
4. Create a variable called score and assign 50 to it. Print the score, change the score to 75, print the score again, change the score to 90, print the score again. Observe how reassignment works.
5. Imagine you went shopping. Create variables for bread(70/=), milk(60/=) and eggs(55/=). Calculate the total amount you have to spend and store it in a variable. Now assume you have 500/=, calculate the amount you have left after shopping and print the result.

TRY USING COMMENTS IN YOUR PROGRAMS WHERE POSSIBLE.

